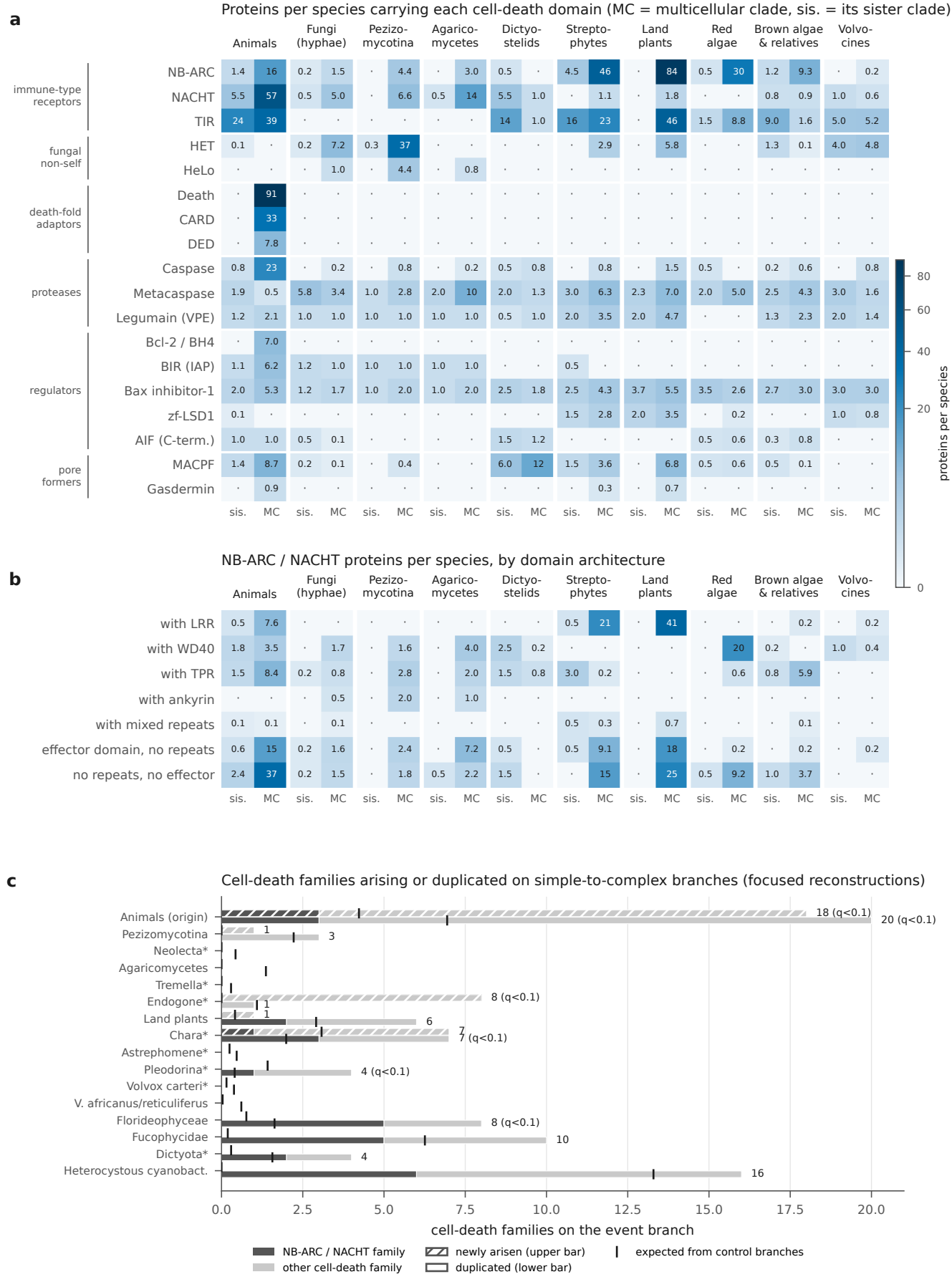
a
Proteins per species carrying each cell-death domain (MC = multicellular clade, sis. = its sister clade)
Animals
Fungi (hyphae)
Pezizo-mycotina
Agarico-mycetes
Dictyo-stelids
Strepto-phytes
Land plants
Red algae
Brown algae & relatives
Volvo-cines
sis.
MC
immune-type receptors
NB-ARC
NACHT
TIR
fungal non-self
HET
HeLo
death-fold adaptors
Death
CARD
DED
proteases
Caspase
Metacaspase
Legumain (VPE)
regulators
Bcl-2 / BH4
BIR (IAP)
Bax inhibitor-1
zf-LSD1
AIF (C-term.)
pore formers
MACPF
Gasdermin
proteins per species
0
20
40
60
80
b
NB-ARC / NACHT proteins per species, by domain architecture
with LRR
with WD40
with TPR
with ankyrin
with mixed repeats
effector domain, no repeats
no repeats, no effector
c
Cell-death families arising or duplicated on simple-to-complex branches (focused reconstructions)
Animals (origin)
Pezizomycotina
Neolecta*
Agaricomycetes
Tremella*
Endogone*
Land plants
Chara*
Astrephomene*
Pleodorina*
Volvox carteri*
V. africanus/reticuliferus
Florideophyceae
Fucophycidae
Dictyota*
Heterocystous cyanobact.
18 (q<0.1)
20 (q<0.1)
8 (q<0.1)
7 (q<0.1)
4 (q<0.1)
8 (q<0.1)
10
16
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
cell-death families on the event branch
NB-ARC / NACHT family
other cell-death family
newly arisen (upper bar)
duplicated (lower bar)
expected from control branches